Title: Project 4 Task 2 Writeup
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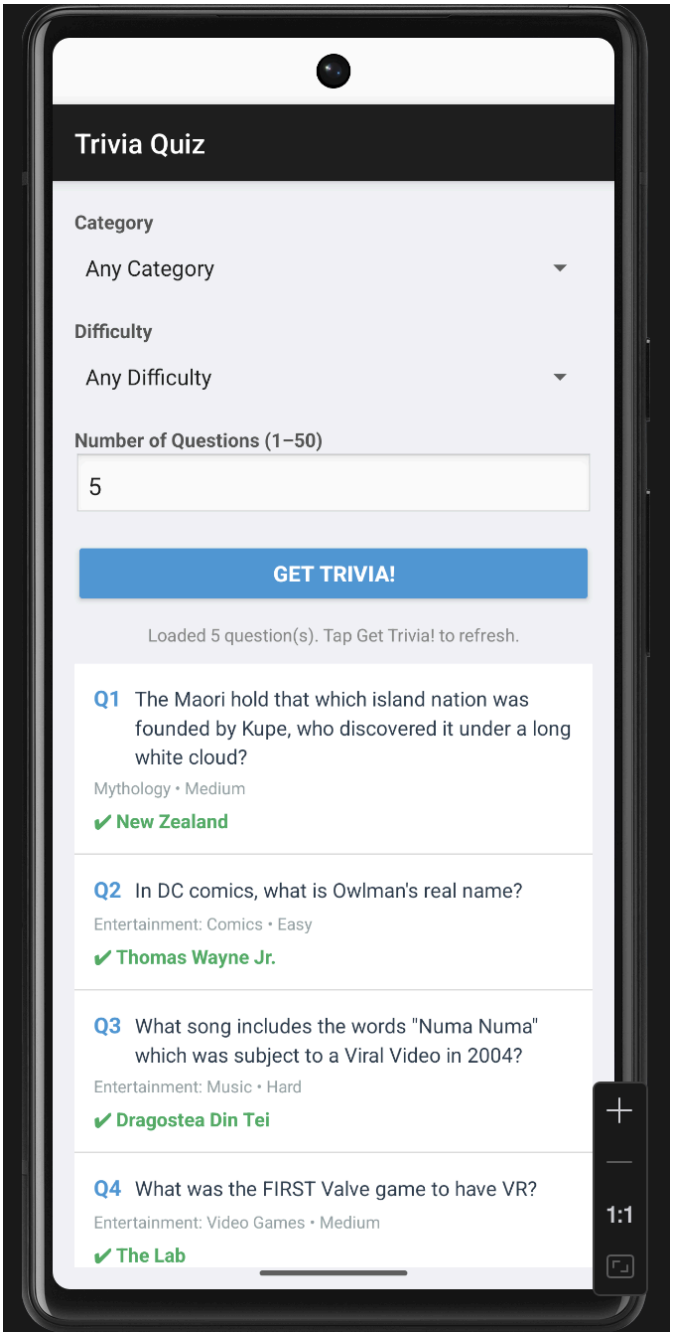
Requirement 1 — Android App

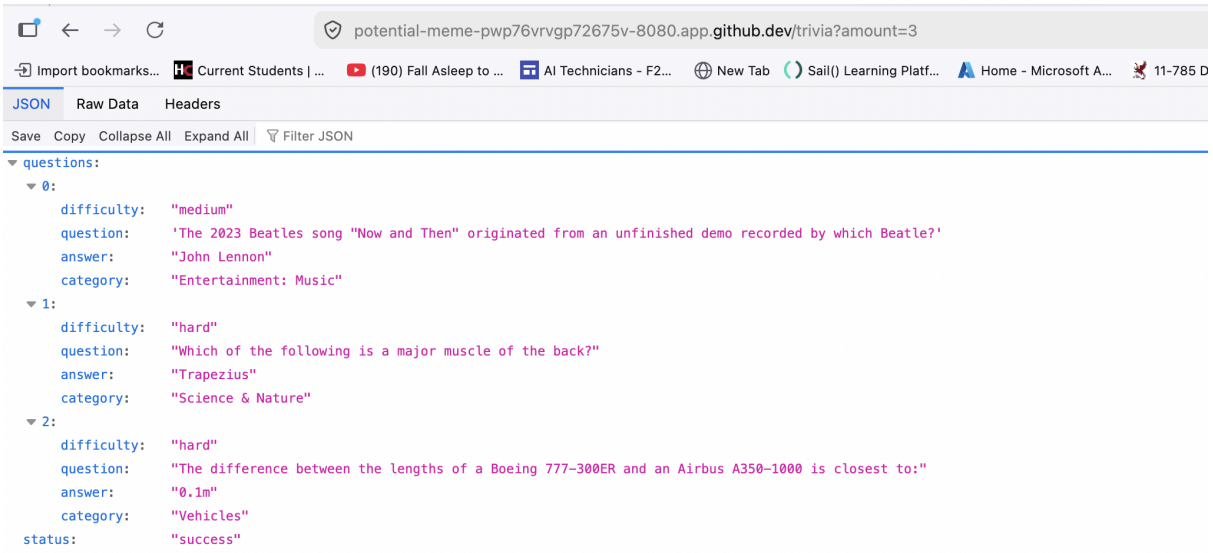
The Android app has 5 different View types: TextView (title, labels, status), Spinner (category and difficulty dropdowns), EditText (question amount input), Button (Get Trivia! button), and ListView (displays results). The user selects a category, difficulty, and amount, then taps Get Trivia! The app sends an HTTP GET request to the web service, parses the JSON response, and displays questions in the ListView. The app is fully repeatable.



Requirement 2 — Web Service

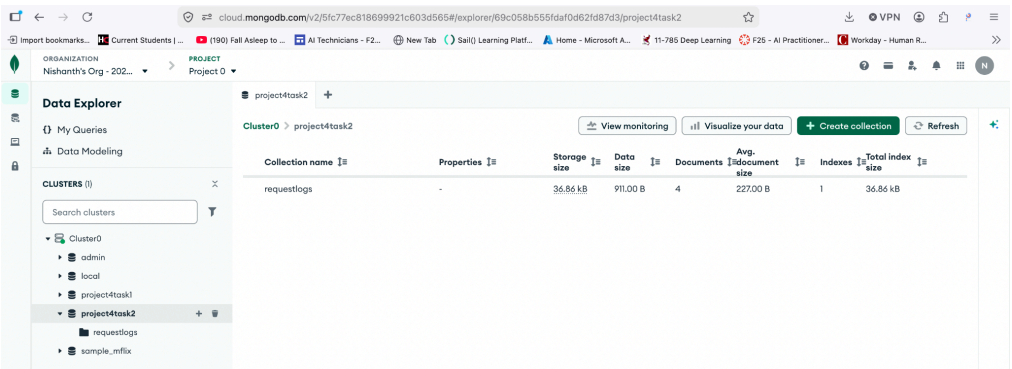
The web service is deployed to GitHub Codespaces at:
https://potential-meme-pwp76vrvgp72675v-8080.app.github.dev/trivia

It receives GET requests from the Android app, fetches data from the OpenTDB API, and returns only the fields the app needs: question, answer, category, difficulty.



Requirement 4 — Logging

8 fields are logged per request to MongoDB: timestamp, deviceModel, ipAddress, category, difficulty, amount, apiResponseTimeMs, questionsReturned.



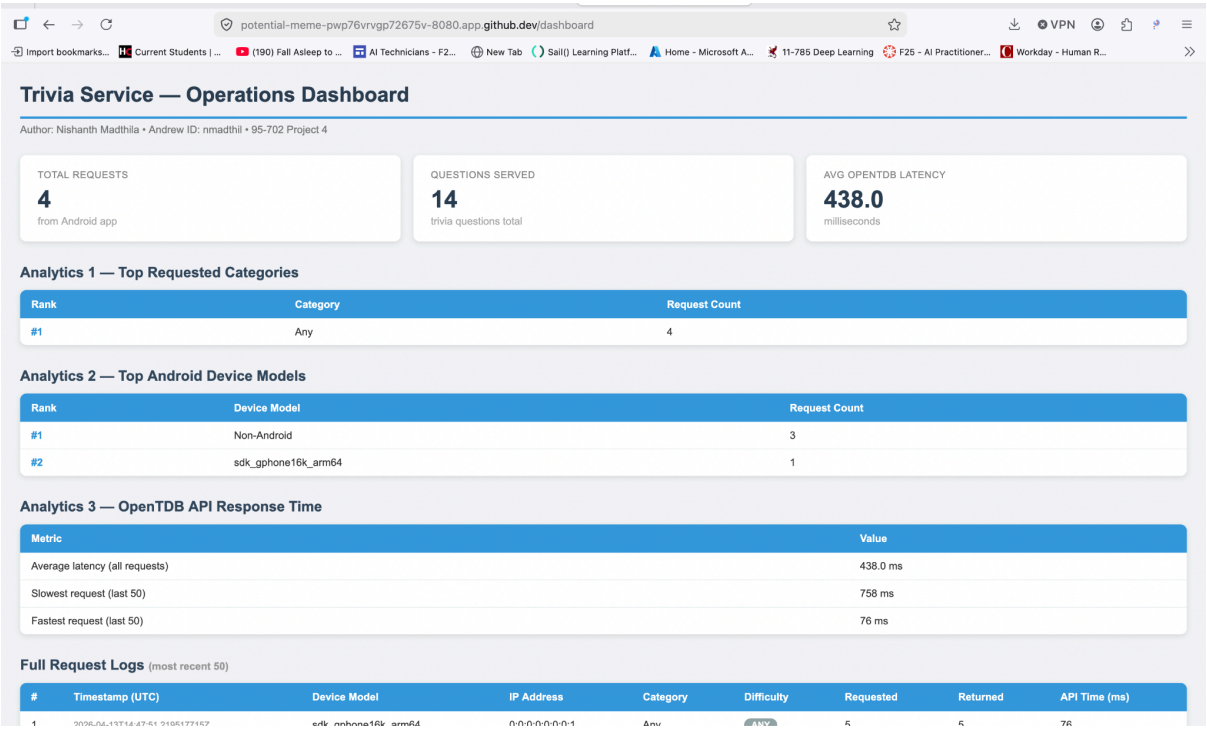
Requirement 5 — MongoDB

Logs are stored in MongoDB Atlas cluster, database project4task2, collection requestlogs.

Requirement 6 — Dashboard

Dashboard URL:
https://potential-meme-pwp76vrvgp72675v-8080.app.github.dev/dashboard

Displays 3 analytics: Top Requested Categories, Top Android Device Models, OpenTDB API Response Time. Also shows full request logs table.



Analytics 1 — Top Requested Categories

Rank	Category	Request Count
#1	Any	4

Analytics 2 — Top Android Device Models

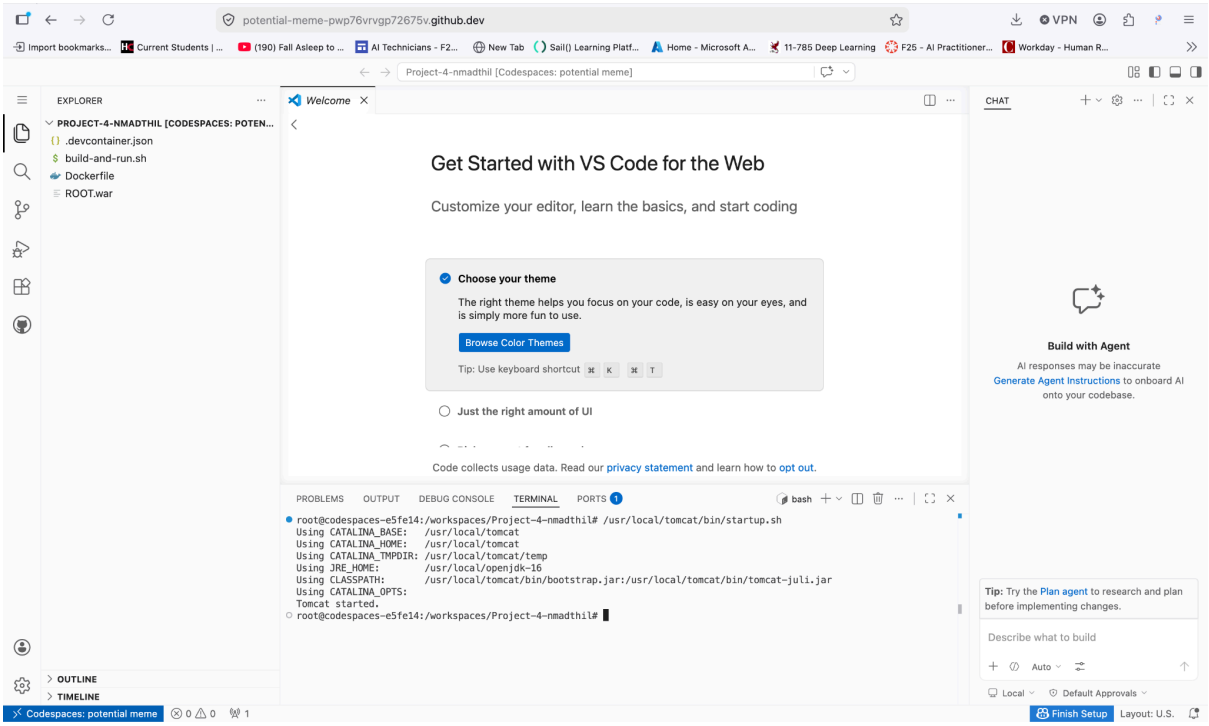
Rank	Device Model	Request Count
#1	Non-Android	3
#2	sdk_gphone16k_arm64	1

Analytics 3 — OpenTDB API Response Time

Metric	Value
Average latency (all requests)	438.0 ms
Slowest request (last 50)	758 ms
Fastest request (last 50)	76 ms

Full Request Logs (most recent 50)

#	Timestamp (UTC)	Device Model	IP Address	Category	Difficulty	Requested	Returned	API Time (ms)
1	2026-04-13T14:47:51.219517715Z	sdk_gphone16k_arm64	0:0:0:0:0:0:0:1	Any	ANY	5	5	76
2	2026-04-13T14:47:36.598672692Z	Non-Android	0:0:0:0:0:0:0:1	Any	ANY	3	3	758
3	2026-04-12T04:30:28.894408811Z	Non-Android	0:0:0:0:0:0:0:1	Any	ANY	3	3	550
4	2026-04-06T22:20:05.764591Z	Non-Android	127.0.0.1	Any	ANY	3	3	368

Requirement 7 — GitHub Codespaces Deployment

Repo: https://github.com/nishsm/Project-4-nmadthil